CSC3034 Computational Intelligence
Tutorial Performance Metrics

1. **Percent correct.** A neural network has been designed to distinguish between four classes, A, B, C, and D. The neural network has two output neurons with the sigmoid activation function. The raw values from the output neurons using the 15 sets of input data are recorded. The expected class of each input data set is provided alongside with the raw values in Table 1.

Table 1: Raw output values and expected classes

Index	Output Neuron 1	Output Neuron 2	Expected Class
1	0.623	0.028	C
2	0.112	0.225	A
3	0.809	0.228	B
4	0.349	0.284	A
5	0.664	0.359	B
6	0.810	0.304	C
7	0.673	0.818	D
8	0.465	0.439	A
9	0.275	0.972	B
10	0.788	0.829	D
11	0.624	0.658	D
12	0.890	0.359	C
13	0.699	0.478	B
14	0.674	0.984	D
15	0.788	0.366	C

To determine the predicted class of a set of input data, the raw values of the output neurons need to be translated into a boolean value using a threshold. A raw value above the threshold is translated to `True`, otherwise, to `False`. The combination of the translated outputs provides the predicted class for the set of input data as shown in Table 2.

Table 2: Predicted class based on translated outputs

Output Neuron 1	Output Neuron 2	Predicted Class
False	False	A
False	True	B
True	False	C
True	True	D

(a) Using the threshold of **0.5**, calculate the percent correct for

i. all the input data (overall percent correct)

Index	Output Neuron 1	Output Neuron 2	Predicted Class	Expected Class
1	0.623	0.028		C
2	0.112	0.225		A
3	0.809	0.228		B
4	0.349	0.284		A
5	0.664	0.359		B
6	0.810	0.304		C
7	0.673	0.818		D
8	0.465	0.439		A
9	0.275	0.972		B
10	0.788	0.829		D
11	0.624	0.658		D
12	0.890	0.359		C
13	0.699	0.478		B
14	0.674	0.984		D
15	0.788	0.366		C

ii. the input data that are supposed to be D

(b) Using the threshold of **0.3**, calculate the percent correct for

i. all the input data (overall percent correct)

Index	Output Neuron 1	Output Neuron 2	Predicted Class	Expected Class
1	0.623	0.028		C
2	0.112	0.225		A
3	0.809	0.228		B
4	0.349	0.284		A
5	0.664	0.359		B
6	0.810	0.304		C
7	0.673	0.818		D
8	0.465	0.439		A
9	0.275	0.972		B
10	0.788	0.829		D
11	0.624	0.658		D
12	0.890	0.359		C
13	0.699	0.478		B
14	0.674	0.984		D
15	0.788	0.366		C

ii. the input data that are supposed to be D

2. **Errors.** Using the data in Question 1, consider `False` as 0 and `True` as 1, identify the expected output of the output neurons for each set of input data.

Index	Output Neuron 1	Output Neuron 2	Expected Class	Expected Output Neuron 1	Expected Output Neuron 2
1	0.623	0.028	C		
2	0.112	0.225	A		
3	0.809	0.228	B		
4	0.349	0.284	A		
5	0.664	0.359	B		
6	0.810	0.304	C		
7	0.673	0.818	D		
8	0.465	0.439	A		
9	0.275	0.972	B		
10	0.788	0.829	D		
11	0.624	0.658	D		
12	0.890	0.359	C		
13	0.699	0.478	B		
14	0.674	0.984	D		
15	0.788	0.366	C		

Note that for this question we will not use the threshold to translate the raw values of the output neurons but just use them as it is.

(a) Calculate the average sum-squared error for the system.

(b) Calculate the mean absolute error for the system.

(c) Calculate the maximum absolute error for the system.

(d) Calculate the normalised error for the system.

3. **Mann-Whitney U test.** In the effort of comparing the performance of GA and PSO, Tim has implemented both GA and PSO to solve the same problem. Both algorithms have been executed for six times each, and the solutions are recorded in Table 3.

Table 3: Solutions provided for the six runs by GA and PSO

Run	1	2	3	4	5	6
GA	-0.031	0.394	0.407	0.389	0.375	0.246
PSO	-0.416	0.263	0.376	0.279	0.042	-0.247

The fitness of the solution can be calculated using Equation 1.

$$f(x) = x^2 + 2x + 2 \tag{1}$$

Using Mann-Whitney U test, evaluate if any of the algorithms perform better than the other. Note that the higher fitness corresponds to better chromosome.

4. **Mann-Whitney U test.** Two evolutionary algorithms are designed to determine the best configuration of a neural network. Daniel intends to compare the performance of the two evolutionary algorithms through the accuracies of the neural network configurations provided.

   EA_1 produces neural networks with accuracies 0.95, 0.89, 0.85, 0.80, and 0.88 whereas EA_2 produces neural networks with accuracies 0.70, 0.93, 0.90, 0.98, 0.82, 0.75, and 0.86.

   Using Mann-Whitney U test to compare the performance of the two evolutionary algorithms.

5. **Confusion Matrices.** Construct the confusion matrix for the data in Table 4.

Table 4: Expected and predicted data

Index	Predicted Class	Expected Class	Index	Predicted Class	Expected Class
1	A	A	26	A	B
2	A	A	27	C	B
3	B	A	28	B	B
4	A	A	29	C	B
5	A	A	30	C	B
6	A	A	31	B	B
7	A	A	32	A	B
8	C	A	33	B	B
9	A	A	34	B	B
10	A	A	35	C	C
11	A	A	36	C	C
12	B	A	37	A	C
13	A	A	38	B	C
14	A	A	39	A	C
15	A	A	40	C	C
16	A	A	41	A	C
17	C	A	42	C	C
18	B	B	43	B	C
19	B	B	44	A	C
20	B	B	45	C	C
21	B	B	46	C	C
22	B	B	47	A	C
23	B	B	48	C	C
24	B	B	49	C	C
25	B	B	50	C	C

		Predicted		
		A	B	C
Actual/Expected	A			
	B			
	C			

(a) Calculate, for the system,

i. the accuracy,

ii. the precision of predicting each classes, and

iii. the recall of predicting each classes.

6. **Confusion Matrices.** An artificial neural network is designed to recognise between three types of vehicles (car, van, bus) from the footage of a CCTV device. The neural network has been tested on 1000 images with 400 images of cars and 300 images of bus.

   The system accuracy is recorded to be 90%. The precision for recognition of van is 0.875. The recalls for the recognition of car and bus are 0.875 and 0.9 respectively. There are 365 images have been recognised as car, and 320 images as van. 5% of the images that contain van are recognised as bus.

   Construct the confusion matrix for the neural network.

7. **Chi-Square test.** A decision tree is used to classify 50 data points into 5 classes, A, B, C, D, and E. The data set contains 10 data points for each class. The result provided by the decision tree classified 5 points to class A, 8 points to B, 15 points to C, 12 points to D, and the rest to E.

   Use chi-square test, with 0.05 level of significance, to comment on the performance of the decision tree.

8. **Chi-Square test.** A neural network is trained to perform classification on the same data set as in question (7). The result of the neural network shows 3 points for class A, 12 points for B, 5 points for C, 18 points for D, and the rest for E.

   Compare the performance of the neural network and the decision tree from (7) using chi-square test at 0.05 level of significance.

CSC3034 Current Course Update

Tutorial Q

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.